

PowerBI Assignment 1 : Data Transformation and Data Modeling

Task 1:

1.Data Import and Transformation:

○ Import List of Orders.csv into Power BI.

The screenshot displays the Power BI Query Editor interface. The main area shows a table with the following columns: Order ID, Order Date, CustomerName, State, and City. The table contains 23 rows of data. The 'APPLIED STEPS' pane on the right shows the steps: Source, Promoted Headers, and Imported List of Order. The 'List of Orders (1)' is highlighted in the 'PROPERTIES' pane.

Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
B-25606	06-04-2018	Hazel	Karnataka	Bangalore
B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
B-25610	09-04-2018	Yogesh	Bihar	Patna
B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
B-25616	15-04-2018	Kanak	Goa	Goa
B-25617	17-04-2018	Sagar	Nagaland	Kohima
B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
B-25620	20-04-2018	Sarita	Maharashtra	Pune
B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
B-25623	22-04-2018	Atharv	West Bengal	Kolkata

The File name called “List of Orders” is imported to the Microsoft Power BI From the Button Get Data→Filename. Text/csv→select and Open the File
And then rename if required, then Click→ Transform Data .

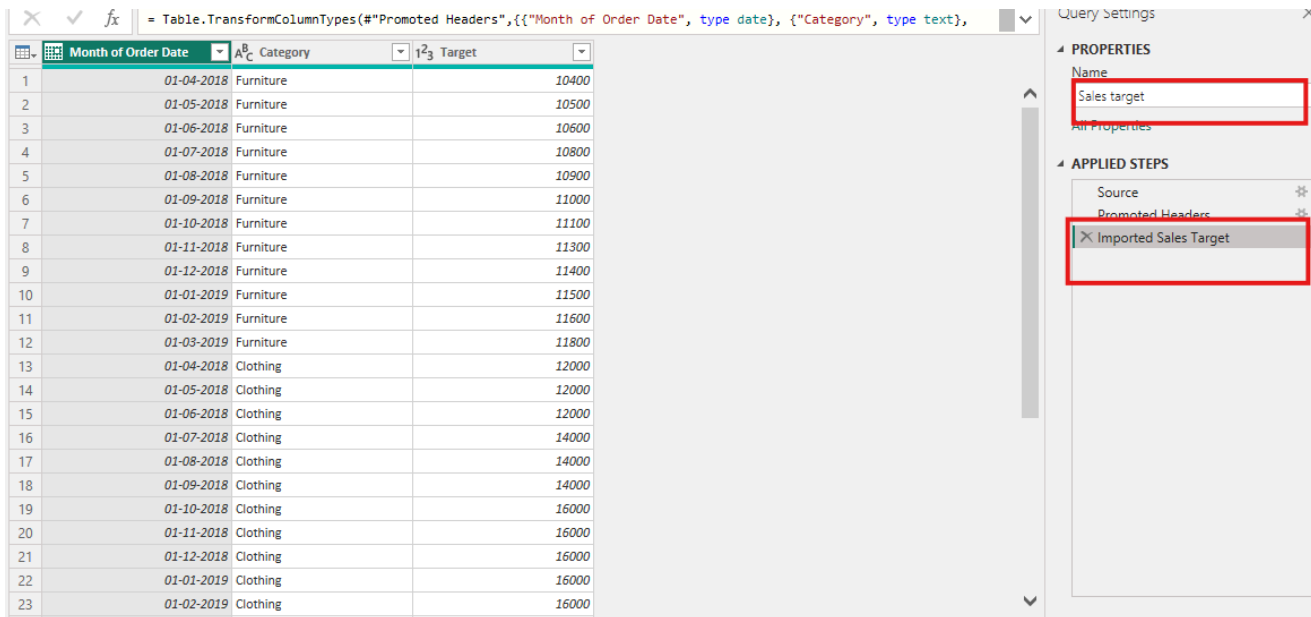
○ Open the list of orders in the Power Query Editor using the Transform Data option.

The screenshot displays the Power BI Query Editor interface. The main area shows a table with the following columns: Order ID, Amount, Profit, Quantity, Category, and Sub-Category. The table contains 23 rows of data. The 'APPLIED STEPS' pane on the right shows the steps: Source, Promoted Headers, and Imported Order Details. The 'Imported Order Details' is highlighted in the 'APPLIED STEPS' pane.

Order ID	Amount	Profit	Quantity	Category	Sub-Category
B-25601	1275	-1148	7	Furniture	Bookcases
B-25601	66	-12	5	Clothing	Stole
B-25601	8	-2	3	Clothing	Hankerchief
B-25601	80	-56	4	Electronics	Electronic G
B-25602	168	-111	2	Electronics	Phones
B-25602	424	-272	5	Electronics	Phones
B-25602	2617	1151	4	Electronics	Phones
B-25602	561	212	3	Clothing	Saree
B-25602	119	-5	8	Clothing	Saree
B-25603	1355	-60	5	Clothing	Trousers
B-25603	24	-30	1	Furniture	Chairs
B-25603	193	-166	3	Clothing	Saree
B-25603	180	5	3	Clothing	Trousers
B-25603	116	16	4	Clothing	Stole
B-25603	107	36	6	Clothing	Stole
B-25603	12	1	2	Clothing	Hankerchief
B-25603	38	18	1	Clothing	Kurti
B-25604	65	17	2	Clothing	T-shirt
B-25604	157	5	9	Clothing	Saree
B-25605	75	0	7	Clothing	Saree
B-25606	87	4	2	Clothing	Shirt
B-25607	50	15	4	Clothing	Leggings
B-25608	1364	-1864	5	Furniture	Tables

In the above image i have imported the Orders Details file using Transform→New source→
 Select Order Details .text/csv →open
 The Order Details table is added in the power Query file.

○ Import Order Details.csv and Sales Target.csv into Power Query Editor:



	Month of Order Date	Category	Target	
1	01-04-2018	Furniture	10400	
2	01-05-2018	Furniture	10500	
3	01-06-2018	Furniture	10600	
4	01-07-2018	Furniture	10800	
5	01-08-2018	Furniture	10900	
6	01-09-2018	Furniture	11000	
7	01-10-2018	Furniture	11100	
8	01-11-2018	Furniture	11300	
9	01-12-2018	Furniture	11400	
10	01-01-2019	Furniture	11500	
11	01-02-2019	Furniture	11600	
12	01-03-2019	Furniture	11800	
13	01-04-2018	Clothing	12000	
14	01-05-2018	Clothing	12000	
15	01-06-2018	Clothing	12000	
16	01-07-2018	Clothing	14000	
17	01-08-2018	Clothing	14000	
18	01-09-2018	Clothing	14000	
19	01-10-2018	Clothing	16000	
20	01-11-2018	Clothing	16000	
21	01-12-2018	Clothing	16000	
22	01-01-2019	Clothing	16000	
23	01-02-2019	Clothing	16000	

From the Transform tab→select New source →Select Delimited as Comma and Click the Transform Data the sales target File is imported to the Power Query.

2. Row Limiting & Data Types:

- Restrict the List of Orders table to the first 500 rows.

fx = Table.FirstN("#Imported List of Order",500)

Order ID	Order Date	CustomerName	State	City
5601	01-04-2018	Bharat	Gujarat	Ahmedabad
5602	01-04-2018	Pearl	Maharashtra	Pune
5603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
5604	03-04-2018	Divsha	Rajasthan	Jaipur
5605	05-04-2018	Kasheen	West Bengal	Kolkata
5606	06-04-2018	Hazel	Karnataka	Bangalore
5607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
5608	08-04-2018	Aarushi	Tamil Nadu	Chennai
5609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
5610	09-04-2018	Yogesh	Bihar	Patna
5611	11-04-2018	Anita	Kerala	Thiruvananthapuram
5612	12-04-2018	Shrichand	Punjab	Chandigarh
5613	12-04-2018	Mukesh	Haryana	Chandigarh
5614	13-04-2018	Vandana	Himachal Pradesh	Simla
5615	15-04-2018	Bhavna	Sikkim	Gangtok
5616	15-04-2018	Kanak	Goa	Goa
5617	17-04-2018	Sagar	Nagaland	Kohima
5618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
5619	18-04-2018	Ramesh	Gujarat	Ahmedabad
5620	20-04-2018	Sarita	Maharashtra	Pune
5621	20-04-2018	Deepak	Madhya Pradesh	Bhopal

Query Settings

PROPERTIES

Name

List of Orders

APPLIED STEPS

Source

Promoted Headers

Imported List of Order

Kept First 500 Rows

In the above Image the Rows of the Customer Name Column has been Reduced to First 500 Rows From the Home Menu—>Select Keep Rows—>Click Keep Top Rows—>Give 500 Rows Click OK.

Then the rows of the Customer Name will be selected as First 500 Rows only.

○ Set the “Order Date column” to the Date data type.

fx = Table.TransformColumnTypes("#Kept First 500 Rows",{{"Order Date", type date}})

Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
B-25606	06-04-2018	Hazel	Karnataka	Bangalore
B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
B-25610	09-04-2018	Yogesh	Bihar	Patna
B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
B-25616	15-04-2018	Kanak	Goa	Goa
B-25617	17-04-2018	Sagar	Nagaland	Kohima
B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
B-25620	20-04-2018	Sarita	Maharashtra	Pune
B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
B-25623	22-04-2018	Atharv	West Bengal	Kolkata

Query Settings

PROPERTIES

Name

List of Orders

APPLIED STEPS

Source

Promoted Headers

Imported List of Order

Kept First 500 Rows

Changed Type -OD-Date

In the Power Query The Table Name is called as “List of Orders” In that table the Column name “Order Date” Data type is converted to Date using Transform Tab→ Choose Data type as Date.

○ **Change the “Amount” and “Target” columns to fixed decimal numbers.**

The screenshot shows the Power Query Editor interface. The main area displays a table with columns: Order ID, Amount, Profit, Quantity, Category, and Sub-Category. The 'Amount' column is highlighted with a red box. The formula bar at the top shows the M code: `= Table.TransformColumnTypes(#"Promoted Headers",{{"Amount", Currency.Type}})`. On the right, the 'Query Settings' pane is open, showing the 'APPLIED STEPS' list. The step 'Amount Changed type \$' is highlighted with a red box.

	Order ID	Amount	Profit	Quantity	Category	Sub-Category
1	B-25601	1,275.00	-148.00	7	Furniture	Bookcases
2	B-25601	66.00	-2.00	5	Clothing	Stole
3	B-25601	8.00	-2.00	3	Clothing	Hankerchief
4	B-25601	80.00	-6.00	4	Electronics	Electronic G
5	B-25602	168.00	-11.00	2	Electronics	Phones
6	B-25602	424.00	-72.00	5	Electronics	Phones
7	B-25602	2,617.00	1151.00	4	Electronics	Phones
8	B-25602	561.00	212.00	3	Clothing	Saree
9	B-25602	119.00	-5.00	8	Clothing	Saree
10	B-25603	1,355.00	-50.00	5	Clothing	Trousers
11	B-25603	24.00	-40.00	1	Furniture	Chairs
12	B-25603	193.00	-66.00	3	Clothing	Saree
13	B-25603	180.00	9.00	3	Clothing	Trousers
14	B-25603	116.00	35.00	4	Clothing	Stole
15	B-25603	107.00	35.00	6	Clothing	Stole
16	B-25603	12.00	1.00	2	Clothing	Hankerchief
17	B-25603	38.00	28.00	1	Clothing	Kurti
18	B-25604	65.00	37.00	2	Clothing	T-shirt
19	B-25604	157.00	9.00	9	Clothing	Saree
20	B-25605	75.00	0.00	7	Clothing	Saree
21	B-25606	87.00	4.00	2	Clothing	Shirt
22	B-25607	50.00	35.00	4	Clothing	Leggings
23	B-25608	1,364.00	-864.00	5	Furniture	Tahlae

In the “Order Details” Table The Column name “Amount” values are converted to Fixed Decimal Number by selecting the Entire Amount Column and then Transform Ribbon it contains →Any Column in that →Data Type Present →Clicking that choose the Option as Fixed Decimal Number then the values are changed to Fixed Decimal Number(\$).

fx = Table.TransformColumnTypes(#"Imported Sales Target",{{"Target", Currency.Type}})

	Month of Order Date	Category	\$ Target
1	01-04-2018	Furniture	10,400.00
2	01-05-2018	Furniture	10,500.00
3	01-06-2018	Furniture	10,600.00
4	01-07-2018	Furniture	10,800.00
5	01-08-2018	Furniture	10,900.00
6	01-09-2018	Furniture	11,000.00
7	01-10-2018	Furniture	11,100.00
8	01-11-2018	Furniture	11,300.00
9	01-12-2018	Furniture	11,400.00
10	01-01-2019	Furniture	11,500.00
11	01-02-2019	Furniture	11,600.00
12	01-03-2019	Furniture	11,800.00
13	01-04-2018	Clothing	12,000.00
14	01-05-2018	Clothing	12,000.00
15	01-06-2018	Clothing	12,000.00
16	01-07-2018	Clothing	14,000.00
17	01-08-2018	Clothing	14,000.00
18	01-09-2018	Clothing	14,000.00
19	01-10-2018	Clothing	16,000.00
20	01-11-2018	Clothing	16,000.00
21	01-12-2018	Clothing	16,000.00
22	01-01-2019	Clothing	16,000.00
23	01-02-2019	Clothing	16,000.00

Query Settings

PROPERTIES

Name
Sales target

All Properties

APPLIED STEPS

Source

Promoted Headers

Imported Sales Target

Target Changed Type \$

Steps for the Target values from the table “Sales target” are similar to the Amount column from the Order Details.

Select the Entire Column of Target Column and In the Column name it contains Data Type → Click that and Select the Data type as Fixed Decimal Number.

3. Text Formatting:

- Format the CustomerName column to proper case to ensure consistent capitalization.

fx = Table.TransformColumns(#"Changed Type -OD-Date",{{"CustomerName", Text.Proper, type text}})

	Order ID	Order Date	CustomerName	State	City
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
15	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
16	B-25616	15-04-2018	Kanak	Goa	Goa
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima
18	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	B-25620	20-04-2018	Sarita	Maharashtra	Pune
21	B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
22	B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
23	B-25623	22-04-2018	Atharv	West Bengal	Kolkata

Query Settings

PROPERTIES

Name

List of Orders

All Properties

APPLIED STEPS

Source

Promoted Headers

Imported List of Order

Kept First 500 Rows

Changed Type -OD-Date

CN-Capitalized Each Word

PREVIEW DOWNLOADED AT 02:3

The Customer Name Column Formatted to Propercase by selecting the full column and right clicking in the Customer name Button in that Choose Capitalized Each Word. Then the Full Customer Name Column will convert to Propercase.

4. Column Creation:

○ Merge the City and State columns to create a new column named Location in the format: City, State

fx = Table.AddColumn(#"CN-Capitalized Each Word", "City,State", each Text.Combine([State], [City], ",", type text))

	Order Date	CustomerName	State	City	City,State
1	01-04-2018	Bharat	Gujarat	Ahmedabad	Gujarat,Ahmedabad
2	01-04-2018	Pearl	Maharashtra	Pune	Maharashtra,Pune
3	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Madhya Pradesh,Bhopal
4	03-04-2018	Divsha	Rajasthan	Jaipur	Rajasthan,Jaipur
5	05-04-2018	Kasheen	West Bengal	Kolkata	West Bengal,Kolkata
6	06-04-2018	Hazel	Karnataka	Bangalore	Karnataka,Bangalore
7	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir	Jammu and Kashmir,Kashmir
8	08-04-2018	Aarushi	Tamil Nadu	Chennai	Tamil Nadu,Chennai
9	09-04-2018	Jitesh	Uttar Pradesh	Lucknow	Uttar Pradesh,Lucknow
10	09-04-2018	Yogesh	Bihar	Patna	Bihar,Patna
11	11-04-2018	Anita	Kerala	Thiruvananthapuram	Kerala,Thiruvananthapuram
12	12-04-2018	Shrichand	Punjab	Chandigarh	Punjab,Chandigarh
13	12-04-2018	Mukesh	Haryana	Chandigarh	Haryana,Chandigarh
14	13-04-2018	Vandana	Himachal Pradesh	Simla	Himachal Pradesh,Simla
15	15-04-2018	Bhavna	Sikkim	Gangtok	Sikkim,Gangtok
16	15-04-2018	Kanak	Goa	Goa	Goa,Goa
17	17-04-2018	Sagar	Nagaland	Kohima	Nagaland,Kohima
18	18-04-2018	Manju	Andhra Pradesh	Hyderabad	Andhra Pradesh,Hyderabad
19	18-04-2018	Ramesh	Gujarat	Ahmedabad	Gujarat,Ahmedabad
20	20-04-2018	Sarita	Maharashtra	Pune	Maharashtra,Pune
21	20-04-2018	Deepak	Madhya Pradesh	Bhopal	Madhya Pradesh,Bhopal
22	22-04-2018	Monisha	Rajasthan	Jaipur	Rajasthan,Jaipur
23	22-04-2018	Atharv	West Bengal	Kolkata	West Bengal,Kolkata

Query Settings

PROPERTIES

Name

List of Orders

All Properties

APPLIED STEPS

Source

Promoted Headers

Imported List of Order

Kept First 500 Rows

Changed Type -OD-Date

CN-Capitalized Each Word

City,State Merged Column

Column City and State is merged by using Add Column→Merge Columns which is merged and created column name as “City,State”.

- **Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage)**

Custom Column

Add a column that is computed from the other columns.

New column name
Profit Margin

Custom column formula ⓘ
= [Profit] / [Amount]

Available columns

- Order ID
- Amount
- Profit
- Quantity
- Category
- Sub-Category

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

From the Add column → select the Custom Column to create the new column as “Profit Margin” using the formula of Profit / Amount.

fx = Table.TransformColumnTypes(#"Added Custom",{"Profit Margin", Percentage.Type})

	Profit	Quantity	Category	Sub-Category	% Profit Margin	
1	1,275.00	-1148	7	Furniture	Bookcases	-90.0
2	66.00	-12	5	Clothing	Stole	-18.1
3	8.00	-2	3	Clothing	Hankerchief	-25.0
4	80.00	-56	4	Electronics	Electronic Games	-70.0
5	168.00	-111	2	Electronics	Phones	-66.0
6	424.00	-272	5	Electronics	Phones	-64.1
7	2,617.00	1151	4	Electronics	Phones	43.9
8	561.00	212	3	Clothing	Saree	37.7
9	119.00	-5	8	Clothing	Saree	-4.2
10	1,355.00	-60	5	Clothing	Trousers	-4.4
11	24.00	-30	1	Furniture	Chairs	-125.0
12	193.00	-166	3	Clothing	Saree	-86.0
13	180.00	5	3	Clothing	Trousers	2.7
14	116.00	16	4	Clothing	Stole	13.7
15	107.00	36	6	Clothing	Stole	33.6
16	12.00	1	2	Clothing	Hankerchief	8.3
17	38.00	18	1	Clothing	Kurti	47.3
18	65.00	17	2	Clothing	T-shirt	26.1
19	157.00	5	9	Clothing	Saree	3.1
20	75.00	0	7	Clothing	Saree	0.0
21	87.00	4	2	Clothing	Shirt	4.6
22	50.00	15	4	Clothing	Leggings	30.0
23	1,364.00	-1854	5	Furniture	Tables	-136.6

Query Settings

PROPERTIES

Name
Order Details

All Properties

APPLIED STEPS

- Source
- Promoted Headers
- Amount Changed type \$
- Changed Type
- Added Custom
- Profit Margin %**

Profit Margin Column has been created using the Custom Column and its converted to Percentage %

5. Conditional Column:

○ Add a conditional column named Profit Status based on:

■ Profit < 0 → Loss ■ Profit = 0 → Break-Even ■ Profit > 0 → Profit

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Profit Status

	Column Name	Operator	Value	Output
If	Profit Margin	is less than	0	Loss
Else if	Profit Margin	equals	0	Break-Even
Else if	Profit Margin	is greater than	0	Profit

Add Clause

Else

OK Cancel

fx = Table.AddColumn(#"Profit Margin %", "Profit Status", each if [Profit Margin] < 0 then "Loss" else if [Profit

	Quantity	Category	Sub-Category	% Profit Margin	Profit Status
1	-1148	7	Furniture	Bookcases	-90.04% Loss
2	-12	5	Clothing	Stole	-18.18% Loss
3	-2	3	Clothing	Hankerchief	-25.00% Loss
4	-56	4	Electronics	Electronic Games	-70.00% Loss
5	-111	2	Electronics	Phones	-66.07% Loss
6	-272	5	Electronics	Phones	-64.15% Loss
7	1151	4	Electronics	Phones	43.98% Profit
8	212	3	Clothing	Saree	37.79% Profit
9	-5	8	Clothing	Saree	-4.20% Loss
10	-60	5	Clothing	Trousers	-4.43% Loss
11	-30	1	Furniture	Chairs	-125.00% Loss
12	-166	3	Clothing	Saree	-86.01% Loss
13	5	3	Clothing	Trousers	2.78% Profit
14	16	4	Clothing	Stole	13.79% Profit
15	36	6	Clothing	Stole	33.64% Profit
16	1	2	Clothing	Hankerchief	8.33% Profit
17	18	1	Clothing	Kurti	47.37% Profit
18	17	2	Clothing	T-shirt	26.15% Profit
19	5	9	Clothing	Saree	3.18% Profit
20	0	7	Clothing	Saree	0.00% Break-Even
21	4	2	Clothing	Shirt	4.60% Profit
22	15	4	Clothing	Leggings	30.00% Profit
23	-1842	5	Furniture	Tables	-135.65% Loss

Query Settings

PROPERTIES

Name

Order Details

All Properties

APPLIED STEPS

Source

Promoted Headers

Amount Changed type \$

Changed Type

Added Custom

Profit Margin %

Profit Status Conditional Colu...

- From the Add Column Tab we have selected the Conditional Column to provide the details of the Profit Margin Details.
- The Condition is first the Profit Margin value is Less than 0 then outcome will be LOSS, the next condition is Profit Margin value is Equal to 0 then the result is Break-Even, and the last Profit Margin is Greater than 0 then result Profit Status will be Profit.

6. Merging Data (Joins):

- Merge the List of Orders and Order Details tables using Order ID.



Merge

Select tables and matching columns to create a merged table.

List of Orders



Order ID	Order Date	CustomerName	State	City	City,State
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Gujarat,Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune	Maharashtra,Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Madhya Pradesh,Bhopal
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur	Rajasthan,Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata	West Bengal,Kolkata

Order Details



Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
B-25601	1,275.00	-1148	7	Furniture	Bookcases	-90.04%	Loss
B-25601	66.00	-12	5	Clothing	Stole	-18.18%	Loss
B-25601	8.00	-2	3	Clothing	Hankerchief	-25.00%	Loss
B-25601	80.00	-56	4	Electronics	Electronic Games	-70.00%	Loss
B-25602	168.00	-111	2	Electronics	Phones	-66.07%	Loss

Join Kind

Full Outer (all rows from both)

☐ Use fuzzy matching to perform the merge

► Fuzzy matching options

✓ The selection matches 500 of 500 rows from the first table, and 1500 of 1...

OK

Cancel

The tables called “List of Orders” and “Orders Details” are merged using Home tab—>Combine—>Merge Queries—>Select Merge Queries as New Table in that Select the First table to be Merged and then select the Common column “Order ID” in that selected Table . Then click the Second table to Merge with the first one ,in that select the Common Column as “Order ID” and select Kind as Full Outer (all rows from both).Then Click “OK” Button.

○ Name the merged output table as Orders Data.

Queries [4]

- List of Orders
- Order Details
- Sales target
- Orders Data

fx = Table.NestedJoin("#List of Orders", {"Order ID"}, #"Order Details", {"Order ID"}, "Order Details",

	Order ID	Order Date	CustomerName	State	City	City,State
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Gujarat,Ahm
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune	Maharashtra
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Madhya Prac
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur	Rajasthan,Jai
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata	West Bengal
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore	Karnataka,Be
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir	Jammu and K
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	Tamil Nadu,C
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow	Uttar Prades
10	B-25610	09-04-2018	Yogesh	Bihar	Patna	Bihar,Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram	Kerala,Thiru
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh	Punjab,Chan
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh	Haryana,Cha
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla	Himachal Pre
15	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok	Sikkim,Gangt
16	B-25616	15-04-2018	Kanak	Goa	Goa	Goa,Goa
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima	Nagaland,Ko
18	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad	Andhra Prad
19	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad	Gujarat,Ahm
20	B-25620	20-04-2018	Sarita	Maharashtra	Pune	Maharashtra
21	B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal	Madhya Prac
22	B-25622	22-04-2018	Monisha	Rajasthan	Jaipur	Rajasthan,Jai
23	B-25623	22-04-2018	Atharu	West Bengal	Kolkata	West Bengal

7 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

Merged data are merged with all the Column in it Likewise “List of Orders”, “Orders Details” as “OrderID”, “Order Date”, “Customer Name”, “State, City”, “City, State”, “Order Details”. And the Created new table is renamed as “Orders Data”.

7. Handling Missing Data & Duplicate Data

● Identify missing values across all tables. ● Define and apply appropriate strategies such as: ○ Replacing missing values ○ Removing records ○ Retaining nulls with justification ● Identify duplicate rows and define a suitable handling strategy based on business logic.

fx = Table.SelectRows(#"City,State Merged Column", each true)

Order ID	Order Date	CustomerName	State	City	City,State
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Gujarat,Ahm
B-25602	01-04-2018	Pearl	Maharashtra	Pune	Maharashtra
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Madhya Prac
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur	Rajasthan,Jai
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata	West Bengal
B-25606	06-04-2018	Hazel	Karnataka	Bangalore	Karnataka,Ba
B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir	Jammu and K
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	Tamil Nadu,C
B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow	Uttar Prades
B-25610	09-04-2018	Yogesh	Bihar	Patna	Bihar,Patna
B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram	Kerala,Thiru
B-25612	12-04-2018	Shrichand	Punjab	Chandigarh	Punjab,Chan
B-25613	12-04-2018	Mukesh	Haryana	Chandigarh	Haryana,Cha
B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla	Himachal Pr
B-25615	15-04-2018	Bhavna	Sikkim	Gangtok	Sikkim,Gangt
B-25616	15-04-2018	Kanak	Goa	Goa	Goa,Goa
B-25617	17-04-2018	Sagar	Nagaland	Kohima	Nagaland,Ko
B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad	Andhra Prad
B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad	Gujarat,Ahm
B-25620	20-04-2018	Sarita	Maharashtra	Pune	Maharashtra
B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal	Madhya Prac
B-25622	22-04-2018	Monisha	Rajasthan	Jaipur	Rajasthan,Jai
B-25623	22-04-2018	Atharv	West Bengal	Kolkata	West Bengal

Query Settings

PROPERTIES

Name
List of Orders

All Properties

APPLIED STEPS

Source

Promoted Headers

Imported List of Order

Kept First 500 Rows

Changed Type -OD-Date

CN-Capitalized Each Word

City,State Merged Column

Filtered Rows

Identifying Missing values for all tables Using Filter Options. Select the particular Column to be Finding missing values in that drop down box we have a filter Select→Remove Empty

fx = Table.TransformColumnTypes(Source,{{"Order Date", type date}})

Order ID	Order Date	CustomerName	State	City	City,State
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Gujarat,Ahm
B-25602	01-04-2018	Pearl	Maharashtra	Pune	Maharashtra
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Madhya Prac
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur	Rajasthan,Jai
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata	West Bengal
B-25606	06-04-2018	Hazel	Karnataka	Bangalore	Karnataka,Ba
B-25607	06-04-2018	Sonakshi	Jammu And Kashmir	Kashmir	Jammu And I
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	Tamil Nadu,C
B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow	Uttar Prades
B-25610	09-04-2018	Yogesh	Bihar	Patna	Bihar,Patna
B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram	Kerala,Thiru
B-25612	12-04-2018	Shrichand	Punjab	Chandigarh	Punjab,Chan
B-25613	12-04-2018	Mukesh	Haryana	Chandigarh	Haryana,Cha
B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla	Himachal Pr
B-25615	15-04-2018	Bhavna	Sikkim	Gangtok	Sikkim,Gangt
B-25616	15-04-2018	Kanak	Goa	Goa	Goa,Goa
B-25617	17-04-2018	Sagar	Nagaland	Kohima	Nagaland,Ko
B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad	Andhra Prad
B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad	Gujarat,Ahm
B-25620	20-04-2018	Sarita	Maharashtra	Pune	Maharashtra
B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal	Madhya Prac
B-25622	22-04-2018	Monisha	Rajasthan	Jaipur	Rajasthan,Jai
B-25623	22-04-2018	Atharv	West Bengal	Kolkata	West Bengal

Query Settings

PROPERTIES

Name
Orders Data

All Properties

APPLIED STEPS

Source

Date Changed type

Removed Duplicates

8. Sorting and Filtering Data

- Sort records in the Orders Data table by Order Date in descending order.

Table Sort("#Removed Duplicates",{"Order Date", Order.Descending}))

	Order ID	Order Date	CustomerName	State	City	City,State
1	B-26100	31-03-2019	Hitika	Madhya Pradesh	Indore	Madhya Prac
2	B-26099	30-03-2019	Bhishm	Maharashtra	Mumbai	Maharashtra
3	B-26098	29-03-2019	Pinky	Jammu And Kashmir	Kashmir	Jammu And I
4	B-26097	28-03-2019	Vini	Karnataka	Bangalore	Karnataka,Be
5	B-26096	28-03-2019	Atharv	West Bengal	Kolkata	West Bengal
6	B-26095	28-03-2019	Monisha	Rajasthan	Jaipur	Rajasthan,Jai
7	B-26094	27-03-2019	Deepak	Madhya Pradesh	Bhopal	Madhya Prad
8	B-26091	27-03-2019	Manju	Andhra Pradesh	Hyderabad	Andhra Prad
9	B-26092	27-03-2019	Ramesh	Gujarat	Ahmedabad	Gujarat,Ahm
10	B-26093	27-03-2019	Sarita	Maharashtra	Pune	Maharashtra
11	B-26090	27-03-2019	Sagar	Nagaland	Kohima	Nagaland,Ko
12	B-26088	26-03-2019	Bhavna	Sikkim	Gangtok	Sikkim,Gangt
13	B-26086	26-03-2019	Mukesh	Haryana	Chandigarh	Haryana,Cha
14	B-26087	26-03-2019	Vandana	Himachal Pradesh	Simla	Himachal Pr
15	B-26085	26-03-2019	Shrichand	Punjab	Chandigarh	Punjab,Chan
16	B-26089	26-03-2019	Kanak	Goa	Goa	Goa,Goa
17	B-26084	25-03-2019	Anita	Kerala	Thiruvananthapuram	Kerala,Thiru
18	B-26083	24-03-2019	Vogesh	Bihar	Patna	Bihar,Patna
19	B-26082	23-03-2019	Itesh	Uttar Pradesh	Lucknow	Uttar Prades
20	B-26078	22-03-2019	Kasheen	West Bengal	Kolkata	West Bengal
21	B-26077	22-03-2019	Divsha	Rajasthan	Jaipur	Rajasthan,Jai
22	B-26080	22-03-2019	Sonakshi	Jammu And Kashmir	Kashmir	Jammu And I
23	B-26079	22-03-2019	Hazel	Karnataka	Bangalore	Karnataka,Be

Query Settings

PROPERTIES

Name

Orders Data

All Properties

APPLIED STEPS

Source

Date Changed type

Removed Duplicates

Sorted Rows-Orders Data/De...

The Column named “Order Date” is Sorted to descending order by using the filter option in that select Descending.

- Apply filters to analyze orders from a specific state (e.g., Tamil Nadu).

Filter Rows

Apply one or more filter conditions to the rows in this table.

Basic Advanced

Keep rows where 'State'

contains

Tamil Nadu

And Or

Enter or select a valu...

OK Cancel

Query Editor Formula Bar: `= Table.SelectRows("#Sorted Rows-Orders Data/Descending", each Text.Contains([State], "Tamil Nadu"))`

	Order ID	Order Date	CustomerName	State	City	City,State
1	26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Tamil Nadu,Chenn
2	26018	14-02-2019	Aarushi	Tamil Nadu	Chennai	Tamil Nadu,Chenn
3	26008	09-02-2019	Kalyani	Tamil Nadu	Chennai	Tamil Nadu,Chenn
4	25860	15-11-2018	Akshay	Tamil Nadu	Chennai	Tamil Nadu,Chenn
5	25788	21-09-2018	Dinesh	Tamil Nadu	Chennai	Tamil Nadu,Chenn
6	25716	11-07-2018	Surabhi	Tamil Nadu	Chennai	Tamil Nadu,Chenn
7	25698	23-06-2018	Amisha	Tamil Nadu	Chennai	Tamil Nadu,Chenn
8	25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	Tamil Nadu,Chenn

Query Settings Panel:

- PROPERTIES: Name: Orders Data
- APPLIED STEPS:
 - Source
 - Date Changed type
 - Removed Duplicates
 - Sorted Rows-Orders Data/De...
 - TN Specific State Orders**

Right clicking in the state column we have an option as GroupBy, using this function we are selecting the particular state to show the result of Specific Orders of Tamilnadu as mentioned in the above image.

9. Grouping and Aggregating Data

- Duplicate the Order Details table and perform the following:
 - Count of Order ID

Query Editor Formula Bar: `List.NonNullCount("#Order Details Removed Duplicates"[Order ID])`

Result: 500

Query Settings Panel:

- PROPERTIES: Name: Duplicate OrdersDetails
- APPLIED STEPS:
 - Source
 - Promoted Headers
 - Amount Changed type \$
 - Changed Type
 - Added Custom
 - Profit Margin %
 - Profit Status Conditional Colu...
 - Order Details Removed Dupli...
 - Calculated Count- Order ID**

In the Power Query , they show Table names with table in left side of the margin Bar.From that Select the Table Name called “Order Details” Table by Right Clicking it select →Duplicate Option then the Duplicate of Order Details will be Create.

Then Calculate Count of the “Order Id” By selecting that Entire Column and Perform the Statistical Calculation in the Transform Tab →Number Column→Select Statistics→Count Values →Then the value is 500 .

○ Average Profit by Category

The screenshot displays the Power Query Editor interface. The formula bar at the top shows the MEX expression: `= Table.Group(#"Order Details Removed Duplicates", {"Category"}, {"Category Profit AVG", each List.Average(...`. Below the formula bar, a preview table is shown with the following data:

	Category	Category Profit AVG
1	Furniture	20.44318182
2	Electronics	-19.37894737
3	Clothing	15.29022082

On the right side, the 'APPLIED STEPS' list is visible, showing the sequence of transformations. The final step, 'Category Profit AVG', is highlighted with a red box.

By Using the Group By Function we have Calculated the Average Profit of Category Column it Contains Furniture,Electronics,Clothing and their Values Respectively 20.44,-19.37,15.29.

○ Total Amount by Sub-Category

fx = Table.RenameColumns("#Grouped Rows",{"Count", "Average Amount"})

Sub-Category	1.2 Average Amount
1 Bookcases	781.8181818
2 Phones	508.3043478
3 Trousers	789.8888889
4 T-shirt	118.7037037
5 Saree	283.031746
6 Shirt	126.75
7 Leggings	44.78947368
8 Tables	991.75
9 Electronic Games	519.2173913
10 Printers	700.9166667
11 Chairs	260.1818182
12 Hankerchief	73.93220339
13 Stole	98.48051948
14 Skirt	32.88
15 Furnishings	148.1034483
16 Accessories	378.88
17 Kurti	64.11111111

Query Settings

PROPERTIES

Name

Duplicate OrdersDetails

All Properties

APPLIED STEPS

Source

Promoted Headers

Amount Changed type \$

Changed Type

Added Custom

Profit Margin %

Profit Status Conditional Colu...

Order Details Removed Dupli...

Grouped Rows

Sub-Cat Average Amount

Sub Category of the Order Details has been calculated using the Group By Function and then the Average Amount of the Sub category is predicted as above.

• Duplicate the Sales Target table and aggregate the total Target by Month of order date.

fx = Table.Group("#Filtered Rows", {"Month of Order Date"}, {"Total Target", each List.Sum([Target]), type nullable

Month of Order Date	1.2 Total Target
1 01-04-2018	31400
2 01-05-2018	31500
3 01-06-2018	31600
4 01-07-2018	33800
5 01-08-2018	33900
6 01-09-2018	34000
7 01-10-2018	36100
8 01-11-2018	36300
9 01-12-2018	36400
10 01-01-2019	43500
11 01-02-2019	43600
12 01-03-2019	43800

Query Settings

PROPERTIES

Name

Duplicate Sales target

All Properties

APPLIED STEPS

Source

Promoted Headers

Imported Sales Target

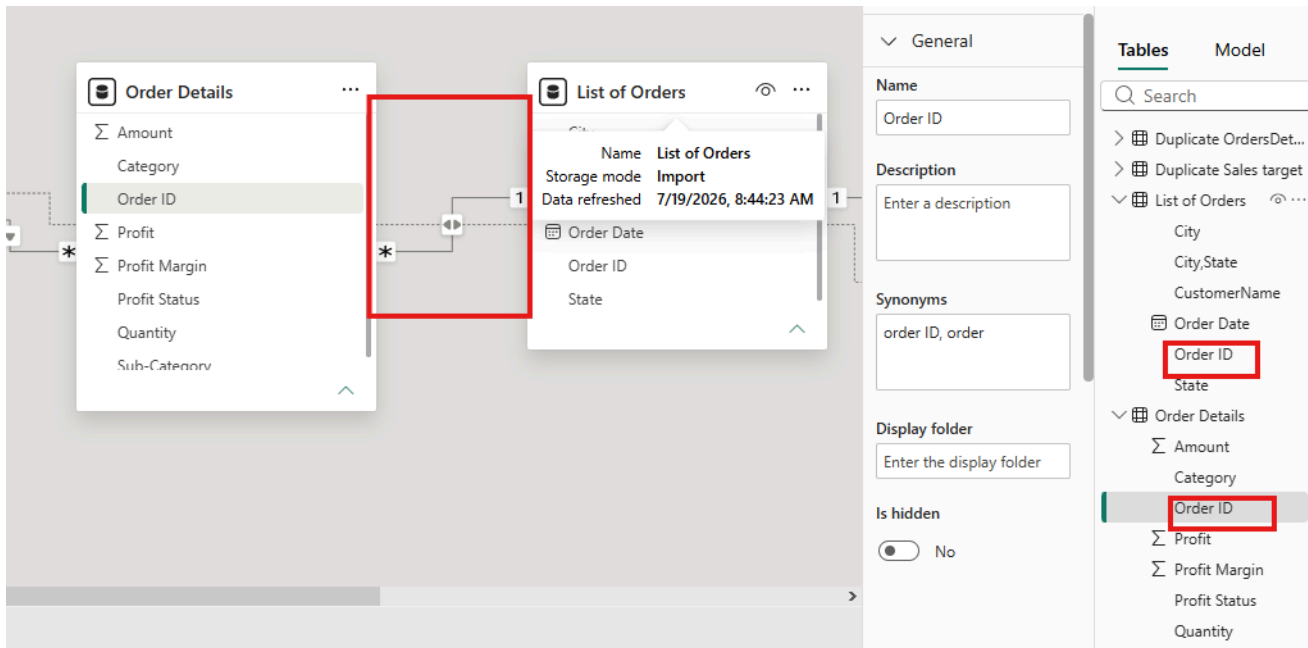
Target Changed Type \$

Filtered Rows

Grouped Rows TT-MODate

10. Data Modeling:

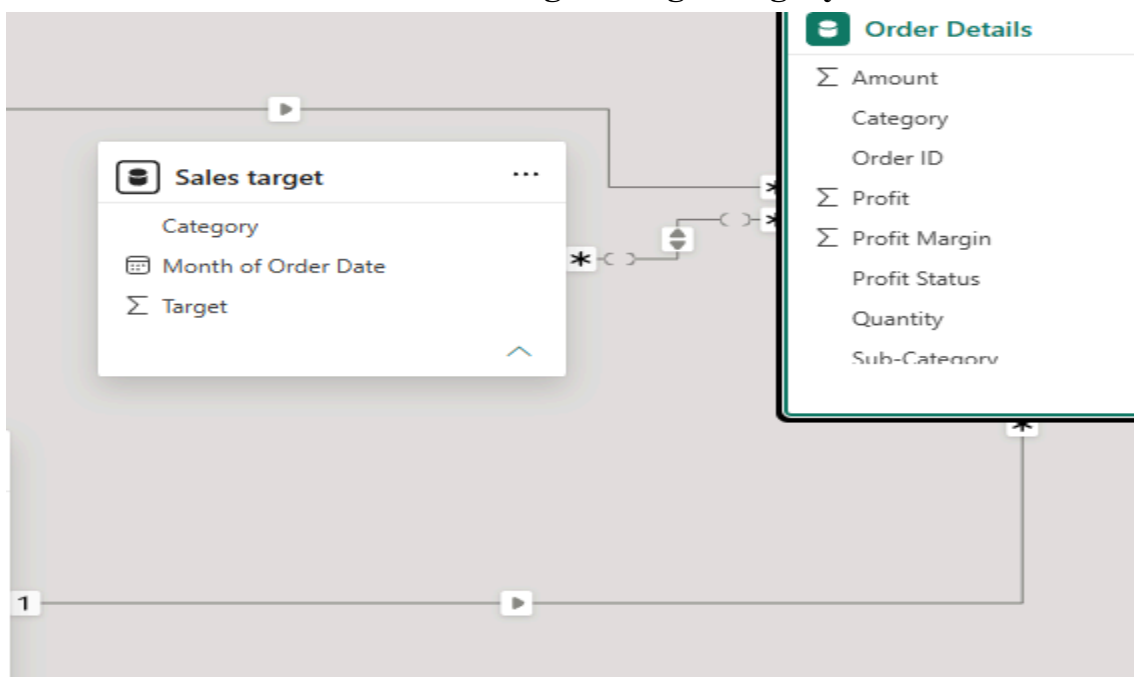
- Establish a relationship between:
 - List of Orders and Order Details using Order ID



The Above image is representing the Establishment of Relationship between “List of Orders” table and “Orders Details”.

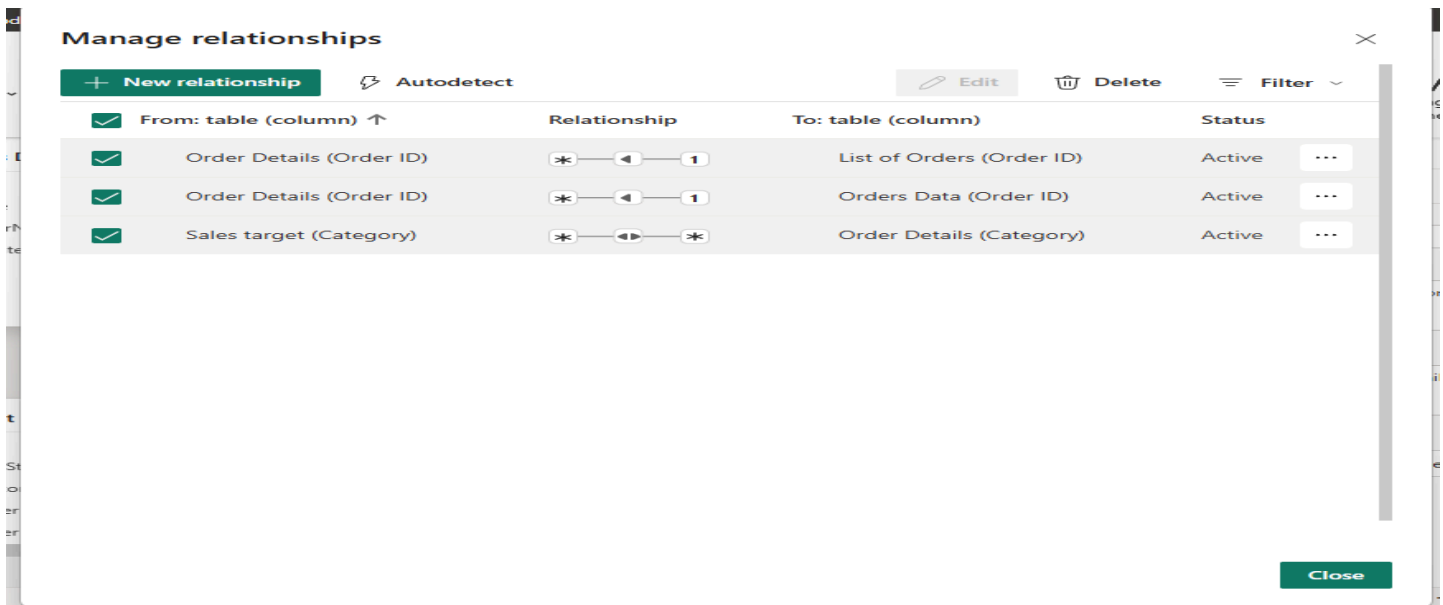
Here the relationship of data modeling is Many to One.

- Establish a relationship between:
 - Order Details and Sales Target using Category



The above Image is representing the Many to many Relationship .Here there are various categories of order that have been filtered in the both tables respectively.

2. Use Manage Relationships to ensure relationships are active and correctly defined.



Here the relationship between the “List of Orders” , “Orders Details”, “Sales Target”, “Orders Data” implemented in the Data Modeling Power BI as Many to One,One to Many,Many to Many.